Compound Inequalities: Solutions & Non-Solutions

(Also available in WeScheme)

Students build upon their understanding of Booleans and simple inequalities to compose compound inequalities using the concepts of union and intersection.

Lesson Goals	Students will be able to: • Understand how the conjunctions `and` and `or` differ • Describe how functions can work together • Describe the solution set of a ***compound inequality***
Student-Facing Lesson Goals	• I can explain the difference between how `and` and `or` are used in programming • I can use two or more inequalities together and describe the area they enclose
Prerequisites	• Simple Data Types • Contracts • Functions: Contracts, Examples & Definitions • Solving Word Problems with the Design Recipe • Simple Inequalities
Materials	• Lesson Slides • Compound Inequalities Starter File • Converting Circles of Evaluation to Code (Page 28) • Compound Inequalities — Practice (Page 29) • Compound Inequalities: Solutions & Non-Solutions (Page 30) • Compound Inequality Functions (Page 31) • Matching Compound Inequality Functions and plots (Desmos) • *Optional: Compound Inequality Warmup*

Glossary

compound inequality :: an inequality that combines two simple inequalities using *and* or *or*

function :: a relation from a set of inputs to a set of possible outputs, where each input is related to exactly one output

intersection :: the set of values that makes both inequalities true

union :: the set of values that makes either or both of a set of inequalities true

Introducing Compound Inequalities

Overview

Students consider the need to *compose* inequalities, and think about how to write them.

Launch

We use inequalities for lots of things:

- Is it hot out? ($temperature > 80^\circ$)
- Did I get paid enough for painting that fence? ($paid \geq \$100$)
- Are the cookies finished baking? ($timer = 0$)

What other examples can you come up with?

Many times we need to *combine* inequalities:

- Should I go to the beach? ($temperature > 80^\circ$ and $weather =$" $sunny$ ")
- Was this burrito worth the price? ($taste =$" $delicious$ " and $price \leq \$15$)

Can you think of examples of when we might want to combine inequalities?

Guide students through other examples of **`and`** and **`or`** with various statements, such as:

- "I'm wearing a red shirt AND I'm a math teacher, true or false?"
- "I'm an NBA basketball star OR I'm having pizza for lunch, true or false?"

This can make for a good sit-down, stand-up activity, where students take turns saying compound Boolean statements and everyone stands if that statement is true for them.

Investigate

Both mathematics and programming have ways of combining - or *composing* - inequalities.

Complete Converting Circles of Evaluation to Code and Compound Inequalities — Practice.

Synthesize

Be really careful to check for understanding here.

- Expressions using **`and`** only produce `true` if both of their sub-expressions are `true`.
- Expressions using **`or`** produce `true` if **either** of their sub-expressions are `true`.

Strategies for English Language Learners

When describing compound inequalities, be careful not to use "English shortcuts". For example, we might say "I am holding a marker *and* an eraser" instead of "I am holding a marker *and* I am holding an eraser." These sentences mean the same thing, but the first one obscures the fact that "and" joins two complete phrases. For ELL/ESL students, this is unecessarily adds to cognitive load!

Solutions and Non-Solutions of Compound Inequalities

Launch

The questions below could be printed from [Compound Inequality Warmup](#).

- What are 4 solutions for $x > 5$?
  - *Possible answers: 5.001, 1000, 24, 73.26*
- What are 4 non-solutions for $x > 5$?
  - *Possible answers: 5, -1000, 4.9, 0*
- What are 4 solutions for $x \leq 15$?
  - *Possible answers: 15, 14.98, -63, 1/2*
- What are 4 non-solutions for $x \leq 15$?
  - *Possible answers: 15.1, 25, 2022, 1000000*
- What numbers are in the solutions set of $x > 5$ **and** $x \leq 15$, making *both* of these inequalities true?
  - *Possible answers: 5.001, 10, 12.374, 15*
- How would that be different from the solution set of $x > 5$ **or** $x \leq 15$, making *at least one* of these inequalities true?
  - *All real numbers are a part of the solution set for $x > 5$ **or** $x \leq 15$ because there aren't any numbers that don't satisfy at least one of these expressions.*

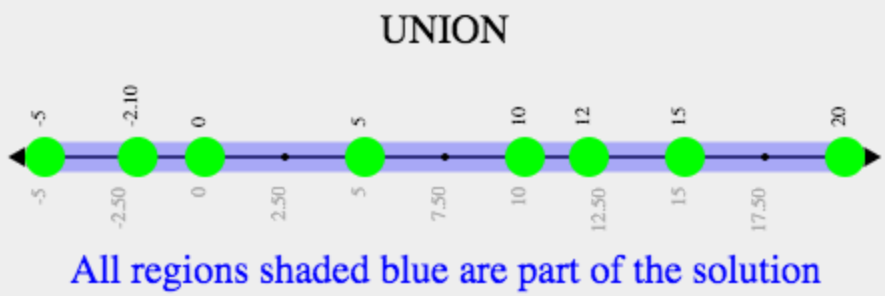
Investigate

- Open the [Compound Inequalities Starter File](#).
- Click "Run" to see the graphs you've just considered.

When students click "Run", four graphs will appear. The top two are the simple inequalities they've just discussed. Encourage students to verify that their solutions and non-solutions are correct. Explain that the bottom two graphs are produced by the special functions `and-intersection` and `or-union`. Use the discussion questions below as a jumping off point in guiding students' analysis of the starter file. *These questions could also be printed from [Exploring Compound Inequality](#).*

- What does `and-intersection` do?
  - *It takes in two functions and a list of numbers and produces a graph with the points and the shaded **intersection** of values that make both of the inequalities true.*

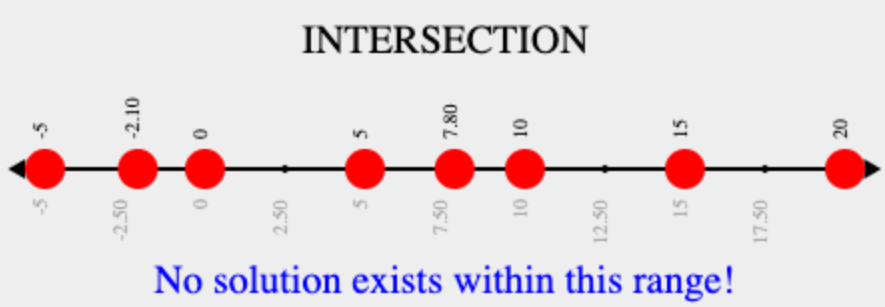
- Why is the circle on 5 red and the circle on 15 green?
  - *Because 5 is not part of the solution - it's not bigger than itself. And 15 is part of the solution - it's less than or equal to 15.*
- Do you think every graph made with `and-intersection` will have different color dots at the ends? Why or why not?
  - *No. It will depend on whether or not the inequality symbols include an equal sign. Sometimes one will and one won't. Sometimes neither will. Sometimes both will.*
- What does `or-union` do?
  - *It takes in two functions and a list of numbers and produces a graph with the points and the shaded* ***union*** *of values that make either or both of the inequalities true.*



- Why did the graph of this `or-union` result in the whole number line being shaded blue?
  - *Because in order to make an* **`or`** *statement true, a number only has to make one of the inequalities true. Every number in the universe is either greater than 5 or less than or equal to 15. There aren't any non-solutions!*
- Not all graphs of `or-union` will look like this. Can you think of a pair of inequalities whose union won't shade the whole graph?
  - *Answers will vary!* $x < 1$ *or* $x < 3$

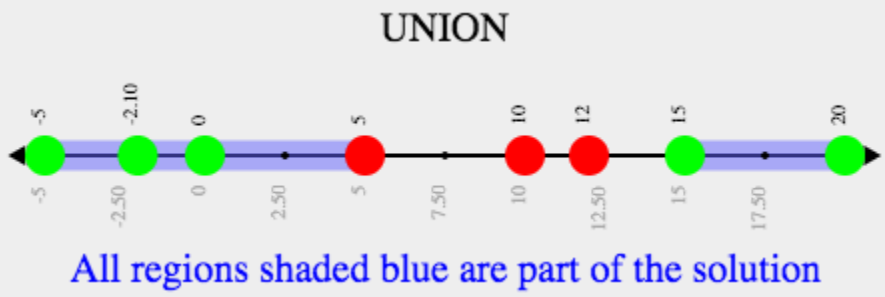
Change the function definition on *line 8* to $x < 5$ and the definition on *line 9* to $x \geq 15$ and, before you click "Run", take a moment to think about what the new graphs of `and-intersection` and `or-union` will look like. Then click "Run" and take a look.

- What does the new `and-intersection` graph look like?
  - *We see a graph with only red circles, representing non-solutions, as below. None of the graph will be shaded blue, because there aren't any numbers that are both smaller than 5 and greater than or equal to 15. A message will tell us that no solutions exist in the provided range.*



- What does the new `or-union` graph look like?

- *There's an arrow to the left and an arrow to the right with a break in the middle between 5 and 15! The dot for 5 is red because it's not part of the solution. The dot for 15 is green because it's part of the solution.*



- Why is the dot for 5 red and the dot for 15 green?
  - *The dot for 5 is red because it's not part of the solution. The dot for 15 is green because it's part of the solution.*
- Which of the 8 numbers from the list are part of the solution set? How do you know?
  - *-5, -2.1, 0, 15, 20… the circles are green.*
- Is 3 part of the solution set? How do you know?
  - *Yes. It's in the blue shaded region.*
- Is 10 part of the solution set? How do you know?
  - *No. It's in the unshaded area.*

Once students are familiar with the starter file, they are ready to use it as they practice identifying solutions and non-solutions for compound inequalities.

- Turn to Compound Inequalities: Solutions & Non-Solutions.
- Explore the compound inequalities listed using the Compound Inequalities Starter File, identifying solutions and non-solutions for each.

Explain to students that instead of defining two functions as simple inequalities, we can produce an inequality graph by defining one function to be a ***compound inequality***! In the following activity, students will analyze inequality graphs and define a *single* function that produces the graph. Walk students through the completed first example before they attempt to write this code on their own.

- Turn to Compound Inequality Functions.
- Write code to describe the compound inequalities pictured.

If you have time, have students open to Matching Compound Inequality Functions and plots (Desmos)

Synthesize

- How did the graphs of intersections and unions differ?

Additional Exercises:

- Converting Circles of Evaluation with Booleans to Code 2